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import cv2
import face_recognition
import numpy as np
import os
from datetime import datetime

# Folders
DATASET_PATH = "dataset"
ATTENDANCE_PATH = "attendance"

if not os.path.exists(DATASET_PATH):
    os.makedirs(DATASET_PATH)

if not os.path.exists(ATTENDANCE_PATH):
    os.makedirs(ATTENDANCE_PATH)

attendance_file = ATTENDANCE_PATH + "/attendance.csv"

# Menu
print("===== FACE ATTENDANCE SYSTEM =====")
print("1. Register New Face")
print("2. Take Attendance")
choice = input("Enter your choice (1/2): ")

# ----- REGISTER FACE -----
if choice == "1":
    name = input("Enter Student Name: ")
    person_path = os.path.join(DATASET_PATH, name)

    if not os.path.exists(person_path):
        os.makedirs(person_path)

    cam = cv2.VideoCapture(0)
    count = 0

    print("Press 'S' to capture images")

    while True:
        ret, frame = cam.read()
        cv2.imshow("Register Face", frame)

        key = cv2.waitKey(1)
        if key == ord('s'):
            img_path = f"{person_path}/{count}.jpg"
            cv2.imwrite(img_path, frame)
            print(f"Image {count+1} saved")
            count += 1

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        if count == 5:
            break

    cam.release()
    cv2.destroyAllWindows()
    print("Face Registration Completed ✅")

# ----- TAKE ATTENDANCE -----
elif choice == "2":

    images = []
    classNames = []

    for folder in os.listdir(DATASET_PATH):
        for img in os.listdir(f"{DATASET_PATH}/{folder}"):
            img_path = f"{DATASET_PATH}/{folder}/{img}"
            image = face_recognition.load_image_file(img_path)
            images.append(image)
            classNames.append(folder)

    def findEncodings(images):
        encodeList = []
        for img in images:
            encode = face_recognition.face_encodings(img)[0]
            encodeList.append(encode)
        return encodeList

    encodeListKnown = findEncodings(images)
    print("Encoding Complete")

    def markAttendance(name):
        with open(attendance_file, 'a') as f:
            now = datetime.now()
            date = now.strftime("%d-%m-%Y")
            time = now.strftime("%H:%M:%S")
            f.write(f'\n{name},{date},{time}')

    cam = cv2.VideoCapture(0)

    while True:
        success, img = cam.read()
        imgS = cv2.resize(img, (0,0), None, 0.25, 0.25)
        imgS = cv2.cvtColor(imgS, cv2.COLOR_BGR2RGB)

        facesCurFrame = face_recognition.face_locations(imgS)
        encodesCurFrame = face_recognition.face_encodings(imgS, facesCurFrame)

        for encodeFace, faceLoc in zip(encodesCurFrame, facesCurFrame):

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matches = face_recognition.compare_faces(encodeListKnown, encodeFace)
faceDis = face_recognition.face_distance(encodeListKnown, encodeFace)
matchIndex = np.argmin(faceDis)

if matches[matchIndex]:
    name = classNames[matchIndex].upper()
    markAttendance(name)

    y1, x2, y2, x1 = faceLoc
    y1, x2, y2, x1 = y1*4, x2*4, y2*4, x1*4
    cv2.rectangle(img, (x1,y1), (x2,y2), (0,255,0), 2)
    cv2.putText(img, name, (x1, y1-10),
                cv2.FONT_HERSHEY_SIMPLEX, 1, (255,0,0), 2)

cv2.imshow("Attendance System", img)
if cv2.waitKey(1) & 0xFF == ord('q'):
    break

cam.release()
cv2.destroyAllWindows()

else:
    print("Invalid Choice ❌")

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